

Mohamed Eldagla

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Education

German International University in Cairo (GIU)

Master's in Data Science, Minor in Software Engineering
Advised by Dr. Caroline Sabty

October 2025 – Present

New Administrative Capital, Egypt

- Featured Coursework: Analytics on the Cloud, Computer Graphics and Visualizations, Data Engineering and Visualizations, Software Engineering Studio

Egypt-Japan University of Science and Technology (E-JUST)

BSc. in Computer Science (AI and Data Science)

2021 – 2025

Alexandria, Egypt

- Featured Coursework: Advanced NLP (CMU), Natural Language Processing, CS 188: Introduction to AI (UC Berkeley), Computer Vision, Neural Networks, New Trends in AI, Mathematics of Data Science, Probability & Statistics

Skills

Languages: Python, C/C++, JavaScript | Toolkits: TensorFlow, PyTorch, OpenCV

Data & Tools: HuggingFace Transformers, PEFT/LoRA, vLLM, Spark, Kafka, SQL, MongoDB, Pandas, Git, Docker, L^AT_EX

Research Experience

Fatima Institute for Global AI Research ↗

Pre-doctoral Fellow — Curiosity-Driven Learning in LLM Agents & Multilingual AI Safety

April 2026 – Present

Remote

- Studying curiosity-driven LLM exploration across inference, SFT, and GRPO with Alfonso Amayuelas (Stanford HAI).
- Testing curiosity signals inspired by Jürgen Schmidhuber: surprisal, learning progress, and information gain.
- Auditing multilingual, multi-turn specification-following with Tasmiah Tahsin Mayeesha (University of North Texas).

PRISM Research Fellowship ↗ —

Center for Human-Compatible AI (CHAI), UC Berkeley

Research Fellow — Reward Model Interpretability

June 2026 – Present

Remote

- Testing whether the reward bases learned by LoRe (Meta Research ↗) map to human-interpretable concepts.

MIT AI Alignment ↗

AI Safety Fellow

June 2026 – August 2026

Remote

- Studied alignment, scalable oversight, and mechanistic interpretability.
- Discussed reward misspecification, alignment faking, and AI threat models in weekly reading groups.

Université Paris-Est Créteil (UPEC)

Data Scientist Intern (NLP)

June 2025 – August 2025

Remote

- Built an LLM-based event discovery system with TinyLlama 1.1B over 50K+ records from 100+ sources.
- Applied ensemble knowledge distillation using Jais 13B, Gemma 3, and TunBERT.

Professional Experience

Egyptian Cabinet Information and Decision Support Center

AI Engineer Intern

July 2025 – September 2025

New Administrative Capital, Egypt

- Led development of a Prime Minister-sponsored RAG-based rumor detection system for the “Ask Egypt” platform.
- Built scalable pipelines for automated verification of news content from trusted governmental and media sources.

Publications

Reward Basis Decomposition for Interpreting Preference Heterogeneity. Eldagla, M. I., Onubogu, I., Rane, P., Sharma, H., Freedman, R. Accepted, **NeurIPS 2026 Workshop: Interpretability as Science: Towards Rigorous Evaluation of Mechanistic Interpretability.**

Uncertainty, Not Concealment: A Mechanistic Reappraisal of Sycophancy in Vision-Language Models. Eldagla, M. I., ElShafei, Z. H., Anter, A. M. Accepted, **NeurIPS 2026 Workshop: Medical Reasoning with Vision Language Foundation Models (Med-Reasoner).**

Specification Drift in Multilingual, Multi-Turn LLM Interactions. Eldagla, M. I., Lia, N. J., Osman, A., Mayeesha, T. T. *In preparation.* Audits specification-following across multilingual, multi-turn conversations.

The Curious Agent: Can Intrinsic Curiosity Guide LLM Exploration? Eldagla, M. I., Hammad, A., Amayuelas, A. *In preparation.* Tests curiosity-driven exploration across inference, SFT, and GRPO.

Mokhiamar, M. O., Mahmoud, A., Eldagla, M. I., et al. (2025). **Blockchain-Integrated Multi-Modal LSTM-CNN Fusion for High-Precision Epileptic Seizure Detection from EEG Signals.** *Knowledge-Based Systems*, 323, 113703. [Paper]

Elshamy, A. A., Metwaly, H. A., Eldagla, M. I., et al. (2025). **EEG Spectrogram Augmentation Using Transformer and Quantum-Inspired Generative Models for Epilepsy Classification.** *JAC-ECC 2025*. [Paper]

Teaching Experience & Service

German International University in Cairo (GIU) — Faculty of Informatics

Winter 2025 – Spring 2026

Teaching Assistant

New Administrative Capital, Egypt

- Taught Data Engineering and Visualizations (50 students), Big Data & NoSQL (80), Programming I (120), and Databases (154); led lectures/labs, projects, office hours, and grading.
- Covered data pipelines and visualization, MongoDB, CassandraDB, SparkML, Kafka, Hadoop, Python, SQL, relational algebra, and database design.

Code in Place (CS106A)

April 2026 – July 2026

Section Leader (Mentor) — Stanford University

Remote

- Led weekly sessions for Stanford's free introductory Python course, serving 16,000+ learners worldwide.

Digital Egypt Cubs Initiative (DECI)

July 2023 – December 2024

Instructor — Udacity

Egypt

- Taught Python, data structures, and algorithms to 40 students aged 12–17 in weekly four-hour, activity-based lessons.

Google Developer Student Clubs (DSC)

October 2021 – May 2023

Core Team Member & CS50x Instructor — E-JUST

Alexandria, Egypt

- Taught C++, data structures, and algorithms to 50+ undergraduates; delivered CS50x lectures to 100+ students.

United Nations — COP27

October 2022 – November 2022

Information Communication Technology Specialist

Sharm El-Sheikh

- Supported the COP27 mobile app with MCIT and UN engineering teams.

Honors & Awards

GIU MSc Data Science Fellowship — full tuition for academic excellence (2025).

E-JUST Academic Excellence — GPA 4.00; final-semester rank 1/120; graduated in the top 10% (2025).

UPEC Master's-Level AI Research Internship — selected as 1 of 30 from 1,300+ applicants as an undergraduate (2025).

House of Representatives Academic Excellence Award (2022).

Fatima Institute for Global AI Research Fellowship — 1 of 67 fellows selected from 700+ applicants (2026).

PRISM AI Safety Fellowship — 50 fellows from 800+ applicants (under 6.25% acceptance).

Black in AI Emerging Leaders in AI (ELAI) (2026).

NTA Presidential Leadership Programme — AI, CS, and leadership; alumni include 10+ ministers/governors (2023).

Nasser Military Academy Shield — Strategic Studies & National Security, Batch 98 (2023); youngest recipient; program prepares civilians for leadership roles.

United Nations COP27, Information Communication Technology Specialist — selected at a 0.04% acceptance rate (2022).

Presidency of the Egyptian Cabinet Internship — 2% acceptance rate.

Certified Trainer of Trainers (TOT) in Sustainability, Ministry of Planning and Economic Development (2025).

Be An Ambassador Program Award, Ministry of Planning and Economic Development (2025).

Data Science Internship, DataScience Middle East — 2nd place across the two-month hybrid cohort; SAS partner.

SAS Curiosity Cup, global data science competition — Top 15 of 107 participants (2024).

Qubit X Qubit: Introduction to Quantum Computing, Frontier Technology Institute and MIT (2022).

IEEE Appx Hackathon, Zewail City — ranked Top 10 of 700 (October 2020).